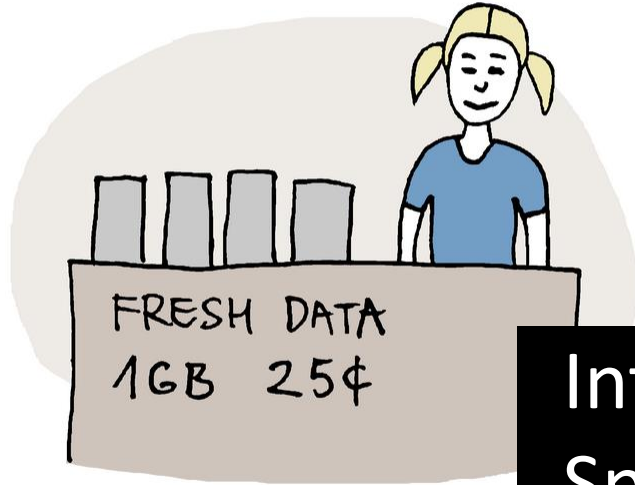


MONETIZING BIG DATA



 Dataedo /cartoon

Introduction to Database Systems Spring 2025, Lecture 12: Big Data Management

Intended Learning Outcomes

1. Suggest a database design in the ER model and convert to a relational database schema in a suitable normal form.
2. Write SQL queries, involving multiple relations, compound conditions, grouping, aggregation, and subqueries.
3. Use relational DBMSs from a conventional programming language in a secure manner.
4. Analyze/predict/improve query processing efficiency of the designed database using indices.
5. Reflect upon the evolution of the hardware and storage hierarchy and its impact on data management system design.
6. Discuss the pros and cons of different classes of data systems for modern analytics and data science applications.

Learning Outcome 6: Discuss the pros and cons of different classes of data systems for modern analytics and data science applications

- **Characteristics of Big Data**
- Classes of Big Data Processing Systems
- Example: MapReduce, Spark

Beyond Small Data

Online Transactional Processing (OTLP)

- Build great interactive applications

➔ “Small” data

This class: beyond small data

- Keep track of all the history
- Keep track of all interactions, also low-level
- Keep track of all data: media, user input, logs

➔ “Big Data”

Characteristics of Big Data

Five Vs of Big Data describe key dimensions of big data

Volume



Velocity



Variety



Veracity



Value





Definition: The amount of data generated, stored and processed.

Significance: Data management system must be scale to given data volume.

Volume

AIRBNB
GUESTS BOOK
747 STAYS

AMAZON
SHOPPERS SPEND
\$455K

X
USERS SEND
360K
TWEETS

6.3M
SEARCHES
HAPPEN ON
GOOGLE

WHATSAPP
USERS SEND
41.6M
MESSAGES

LINKEDIN
USERS SUBMIT
6,060
RESUMES

VIEWERS WATCH
43 YEARS
OF STREAMING
CONTENT

3,720
USERS DOWNLOAD
INSTAGRAM
THREADS

CHATGPT
USERS SEND
6,944 PROMPTS

FACEBOOK
USERS LIKE

EVERY MINUTE
01:00
OF THE DAY

PRESENTED BY

DOMO



CYBER-
CRIMINALS
LAUNCH 30 DDOS ATTACKS

INSTAGRAM
USERS SEND
694K REELS VIA DM

DOORDASH
DINERS PLACE

Rivian electric vehicles generate multiple TB of data per day



CERN: „In June 2022, we had 424 PB of data on tapes”

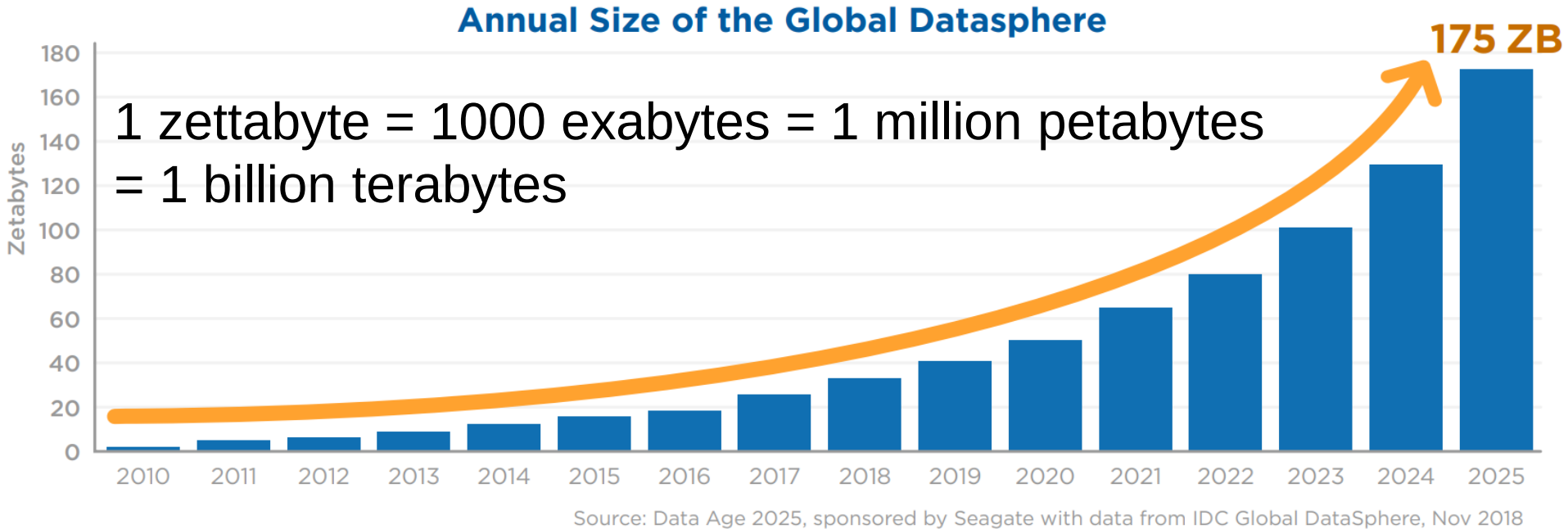


301,722 Datasets Available

15

YEARS OF DATA.GOV

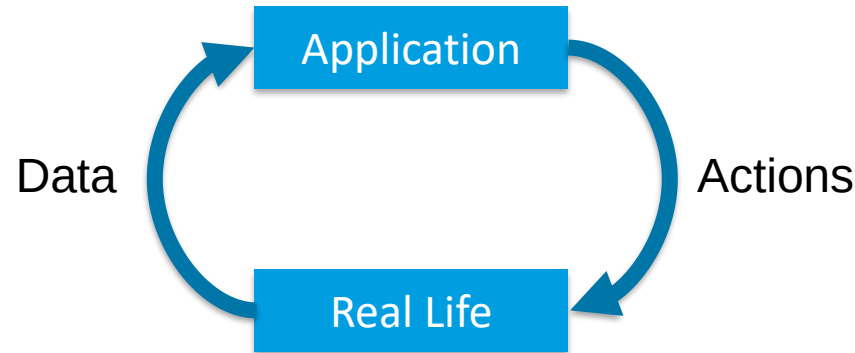
Growth of Data Globally



Source: [Reinsel, Gantz, Rydning; The Digitization of the World from Edge to Core: IDC White Paper 2018](#)



Definition: The speed at which data is generated, collected, and processed.



Significance: Requirement on the system to process data at given speed.



Definition: The diversity of data types and sources. Data can come in **structured, semi-structured, or unstructured** formats.

Significance: Challenge of managing different types of data. Choose data management system for your use case.



Structured data

- Highly regular structure, repeating patterns
- Example: relational data

ID	Name
1	Pinar
2	Veronika
3	Dovile
4	Zoi
5	Martin
6	Eleni



Semi-structured data

- Some structure, changes over time
- Examples: logs, comment threads, graphs

Common Log Format*

```
127.0.0.1 alice Alice [06/May/2021:11:26:42 +0200] "GET / HTTP/1.1" 200 3477
```

Same log line in JSON

```
{  
  "ip_address": "127.0.0.1",  
  "user_id": "alice",  
  "username": "Alice",  
  "timestamp": "06/May/2021:11:26:42 +0200",  
  "request_method": "GET",  
  "request_url": "/",  
  "protocol": "HTTP/1.1",  
  "status_code": 200,  
  "response_size_bytes": 3477  
}
```



PHONES OUT

Advantages of JSON over Common Log Format



Menti.com Code: 1129 2859

Common Log Format

```
127.0.0.1 alice Alice [06/May/2021:11:26:42 +0200] "GET / HTTP/1.1" 200 3477
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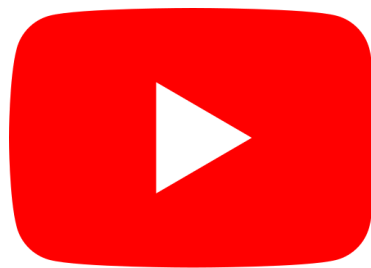
Same log line in JSON

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{  
  "ip_address": "127.0.0.1",  
  "user_id": "alice",  
  "username": "Alice",  
  "timestamp": "06/May/2021:11:26:42 +0200",  
  "request_method": "GET",  
  "request_url": "/",  
  "protocol": "HTTP/1.1",  
  "status_code": 200,  
  "response_size_bytes": 3477  
}
```



Unstructured data

- Little structure
- Examples: photos, videos, music





Definition: The accuracy and trustworthiness of data.

Significance: Data quality varies, and not all collected data can be trusted. It's essential to filter, clean, and validate data to ensure accurate insights.



Only 3% of Companies' Data Meets Basic Quality Standards

by Tadhg Nagle, Thomas C. Redman, and David Sammon

September 11, 2017

Source: [Only 3% of Companies' Data Meets Basic Quality Standards](#), Harvard Business Review



A 2016 study by IBM is even more eye-popping. IBM found that poor data quality strips \$3.1 trillion from the U.S. economy annually due to lower productivity, system outages and higher maintenance costs, to name only a handful of the bad outcomes that result from poor data quality.

Source: [Flying Blind: How Bad Data Undermines Business](#), Forbes



Definition: The accuracy and trustworthiness of data.

Significance: Data quality varies, and not all collected data can be trusted. It's essential to filter, clean, and validate data to ensure accurate insights.



Definition: The ability to turn data into insights for decision making and creating business value.

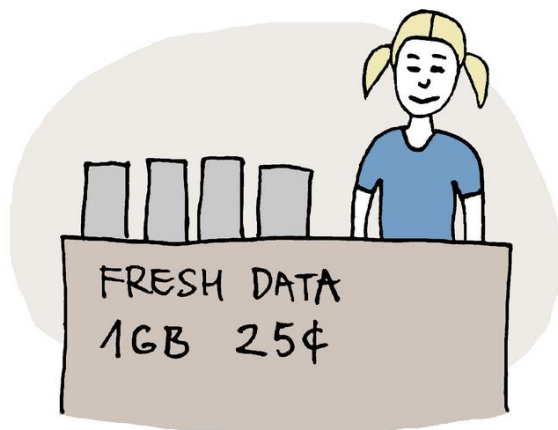
Significance: Big data is only beneficial if it delivers insights.

- Businesses: more effective operations, stronger customer relationships, \$\$\$
- Science: new discoveries
- Public sector: better serving people



Only half a joke: data marketplaces are a thing!

MONETIZING BIG DATA



 Dataedo /cartoon

Plot@Dataedo



Snowflake Data Sharing & Marketplace

DATA SHARING¹

36%

of customers¹ have at ≥ 1 stable edge¹

MARKETPLACE LISTINGS¹

2,946

26% Y/Y Growth

Source: [Snowflake Investor Presentation](#), October 31, 2024

Characteristics of Big Data

Five Vs of Big Data describe key dimensions of big data

Volume



Velocity



Variety



Veracity



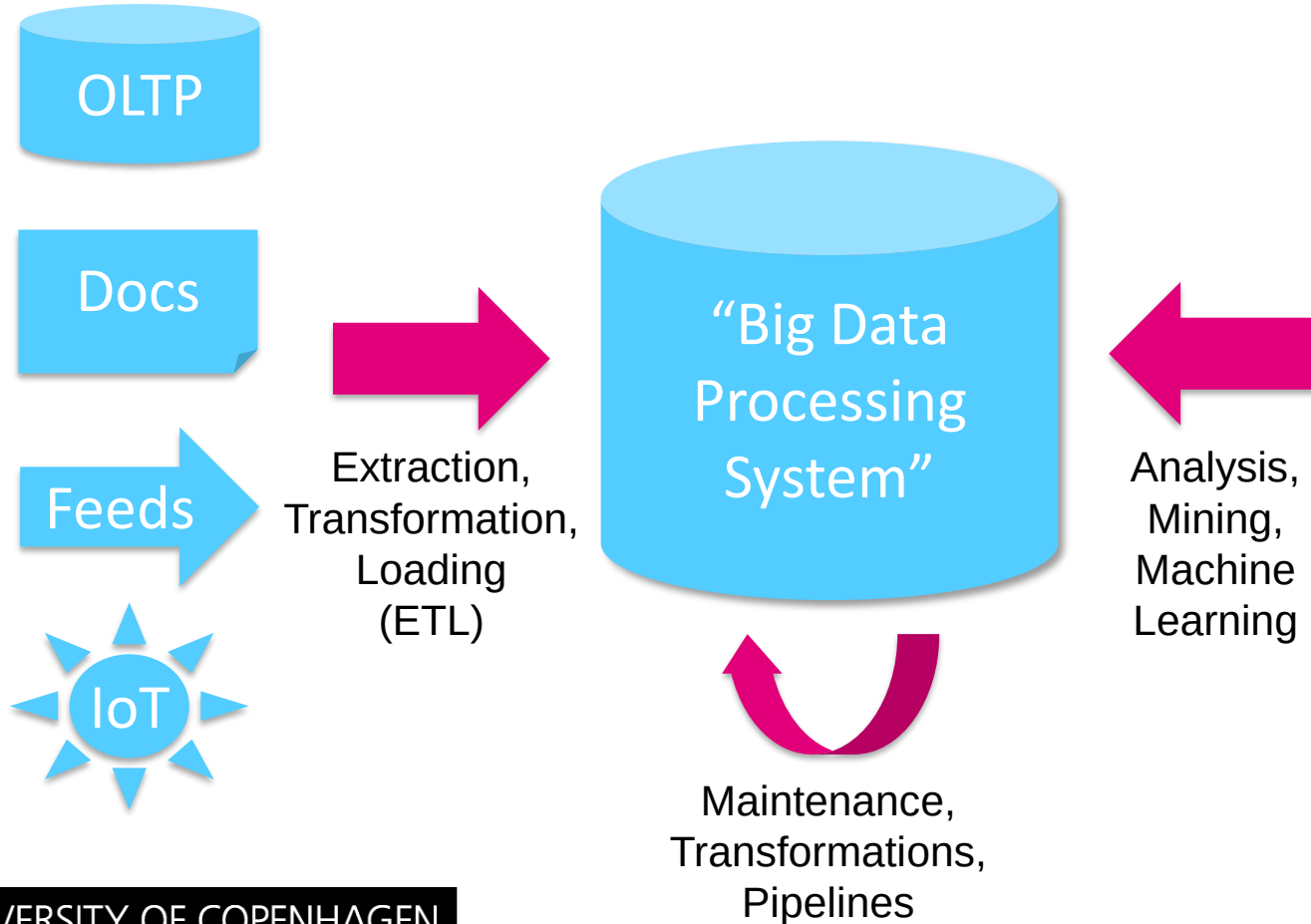
Value



Learning Outcome 6: Discuss the pros and cons of different classes of data systems for modern analytics and data science applications

- Characteristics of Big Data
- **Classes of Big Data Processing Systems**
- Example: MapReduce, Spark

Big Data Processing Overview



Access Patterns to Big Data

Data collections typically consists of many millions of files of data

- Too large for a single server → Distributed storage
- Reading large amounts of data → Sequential scans

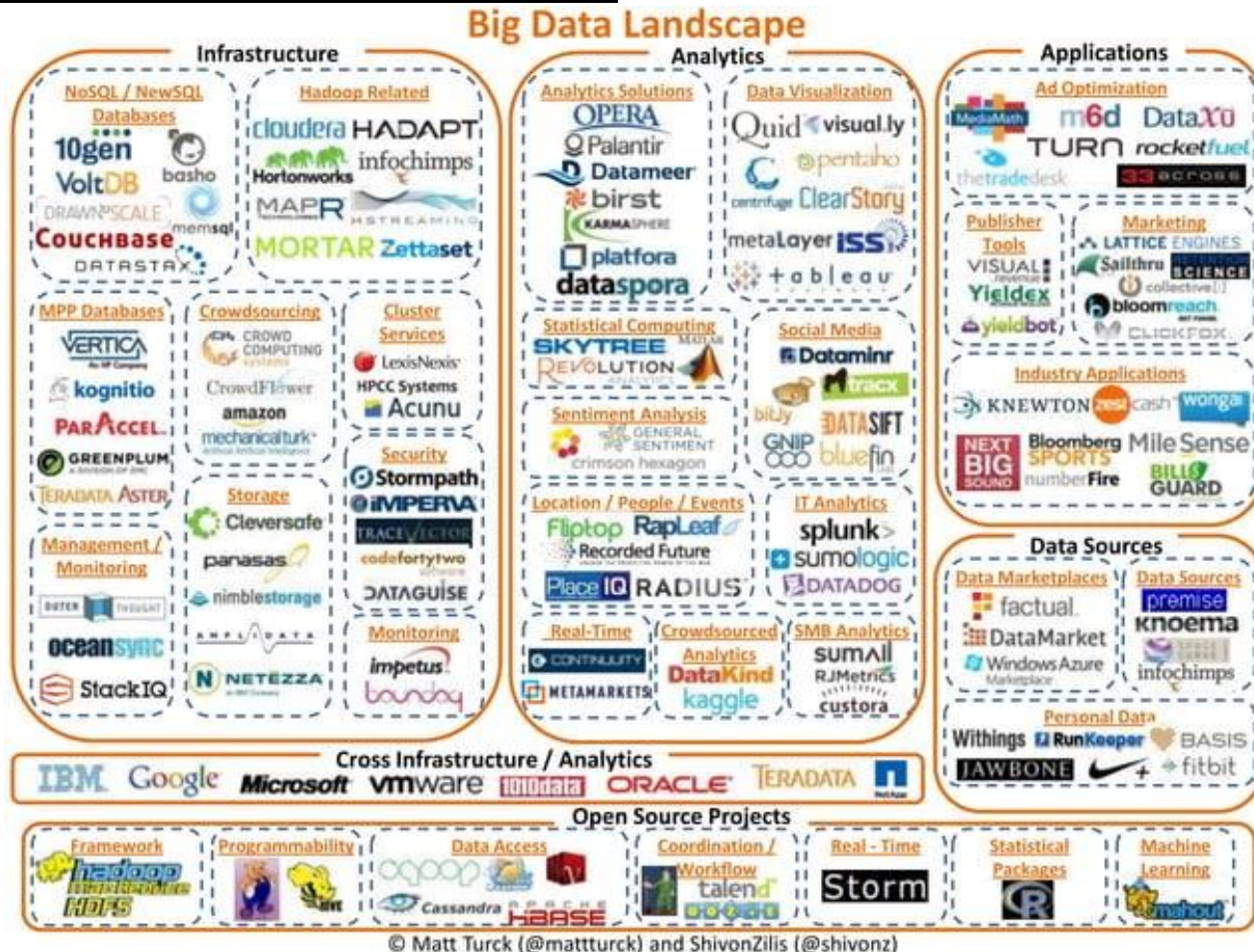
Online Analytical Processing (OLAP)



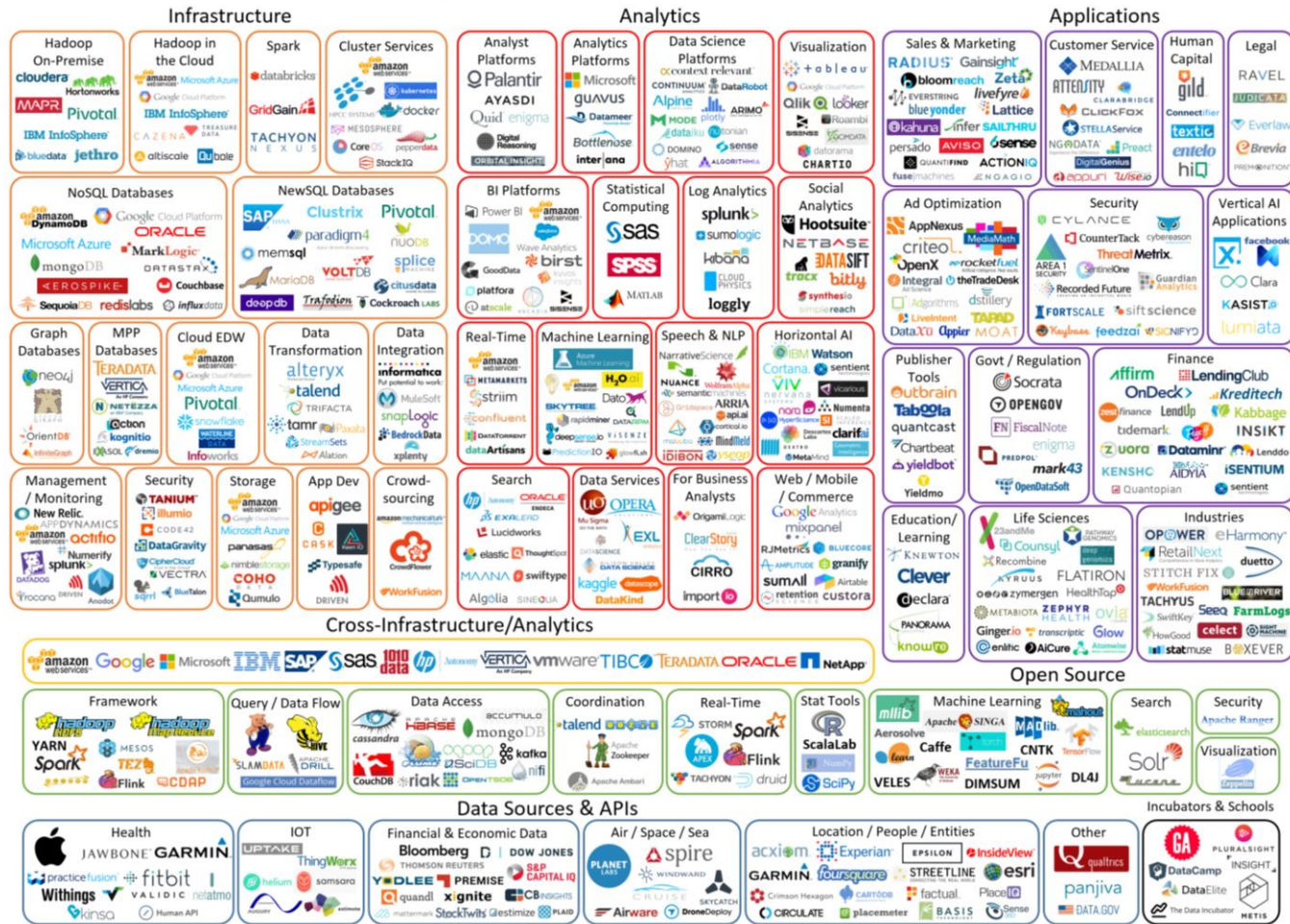
OLAP Database System

- Distributed storage for exabytes of data
- Distributed compute, typically using large, sequential scans
- Large ecosystem of tools for complex data analysis tasks

Big Data Landscape 2012



Big Data Landscape 2016

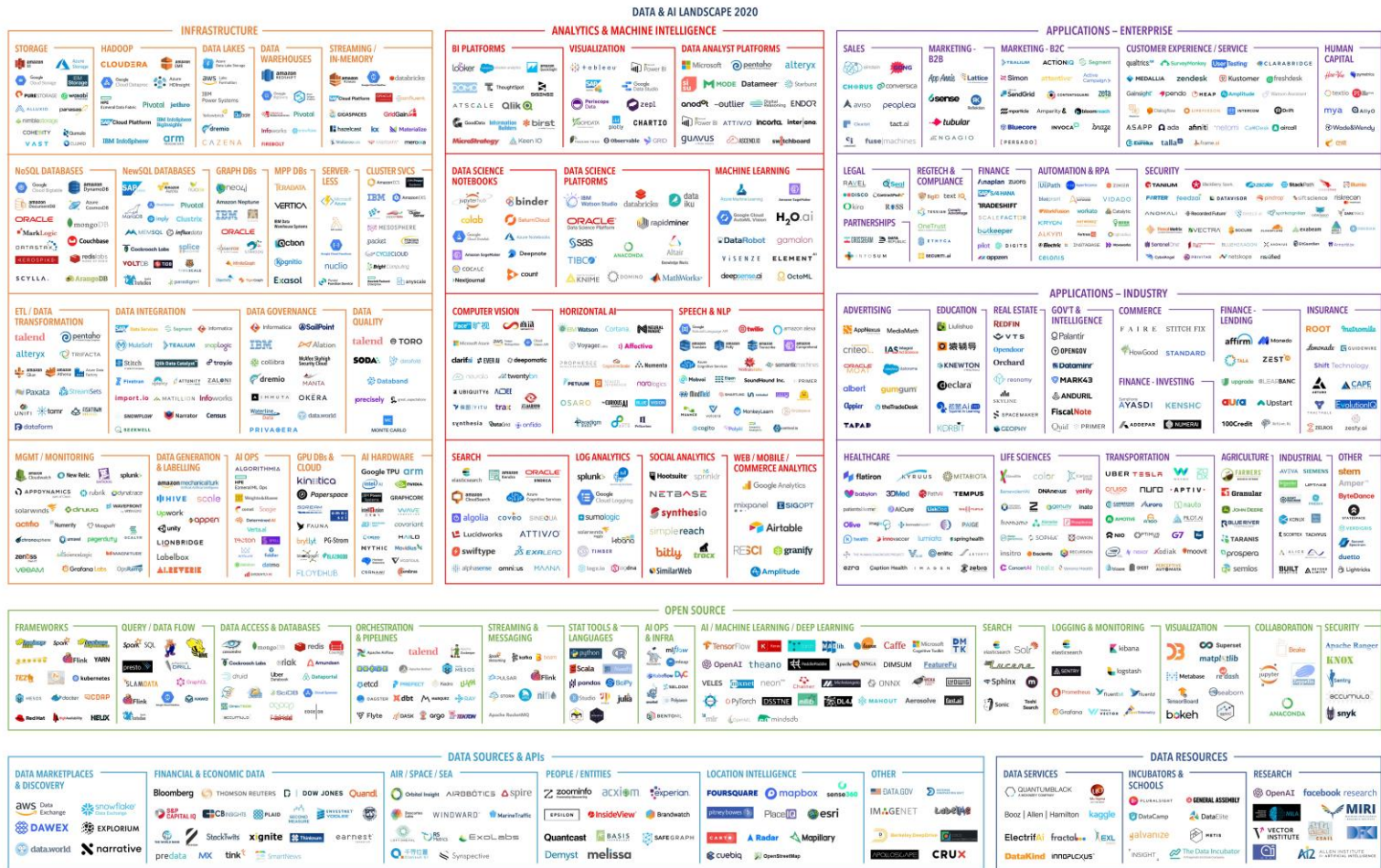


Last Updated 3/23/2016

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FIRST MARK

Big Data Landscape 2020



Version 1.0 - September 2020

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mattturck.com/data2020

FIRSTMARK
EARLY STAGE VENTURE CAPITAL

Big Data Landscape 2023

THE 2023 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE



Conclusions:

- No product does it all
- Typically, you use a set of products
e.g. ETL → Storage & Processing → UI, Reporting, Serving
- Interesting space for software engineers, sales engineers, sales

Version 1.0 - Feb 2023

© Matt Turck (@mattturck), Kevin Zhang (@ykevinzhang) & FirstMark (@firstmarkcap)

Blog post: mattturck.com/MAD2023

Interactive version: MAD.firstmarkcap.com

Comments? Email MAD2023@firstmarkcap.com

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Classes of Big Data Processing Systems

Relational Database Systems

Strengths

- SQL programming API
- Huge SQL ecosystem
- ACID transactions
- Complex query processing

Weaknesses

- Little support for unstructured data
- Little support for machine learning training and serving



Classes of Big Data Processing Systems

MapReduce Systems

Strengths

- Distributed storage
- Great scalability
- Able to process great variety of data sets (structured, semi-structured, unstructured)

Weaknesses

revealed later



CLOUDERA



Classes of Big Data Processing Systems

Key-value Stores

Strengths

- Distributed storage
- Great scalability
- Short latency for single items
- Quick, “out of the box” way to store data

Weaknesses

- Limited storage model (only key, value pairs)
- Limited query interface (no SQL)
- No means to scan and filter data



Classes of Big Data Processing Systems

Document Databases

Strengths

- Store objects / XML / JSON in hierarchical form
- Good integration with object-oriented languages and JavaScript

Weaknesses

- Limited query interface (no SQL)
- No ACID guarantees
- Not designed for scans (very related to key-value stores)



Classes of Big Data Processing Systems

Graph Databases

Strengths

- Capture graph relationships, e.g. knowledge graphs, social networks
- Fast at traversing edge chains, no joins needed



Weaknesses

- Specific to graph applications
- Not many such use cases
- Relational databases outperform graph databases these days



Classes of Big Data Processing Systems

Data Lakehouses

Strengths

- Data storage in open data formats (Apache Parquet + Apache Iceberg) in the cloud
- Great ML support, e.g. training, but also Python-based notebooks

Weaknesses

- Typical API is Apache Spark, less versatile and supported than SQL
- However: Lakehouses are moving to SQL



databricks

Learning Outcome 6: Discuss the pros and cons of different classes of data systems for modern analytics and data science applications

- Characteristics of Big Data
- Classes of Big Data Processing Systems
- **Example: MapReduce**, Spark

MapReduce Overview

Early tools to deal with Big Data

- MapReduce by Google in 2003
- Open-source Apache Hadoop based on MapReduce

Interface

- Map $(k1, v1) \rightarrow \text{list}(k2, v2)$
- Reduce $(k2, \text{list}(v2)) \rightarrow (k2, v3)$

MapReduce Framework

Framework

- Read lots of data (e.g. text documents)
- **Map:** process a data item
- Sort and shuffle
- **Reduce:** aggregate
- Write results

Map and Reduce functions are user-specified

MapReduce Phases

Map Phase

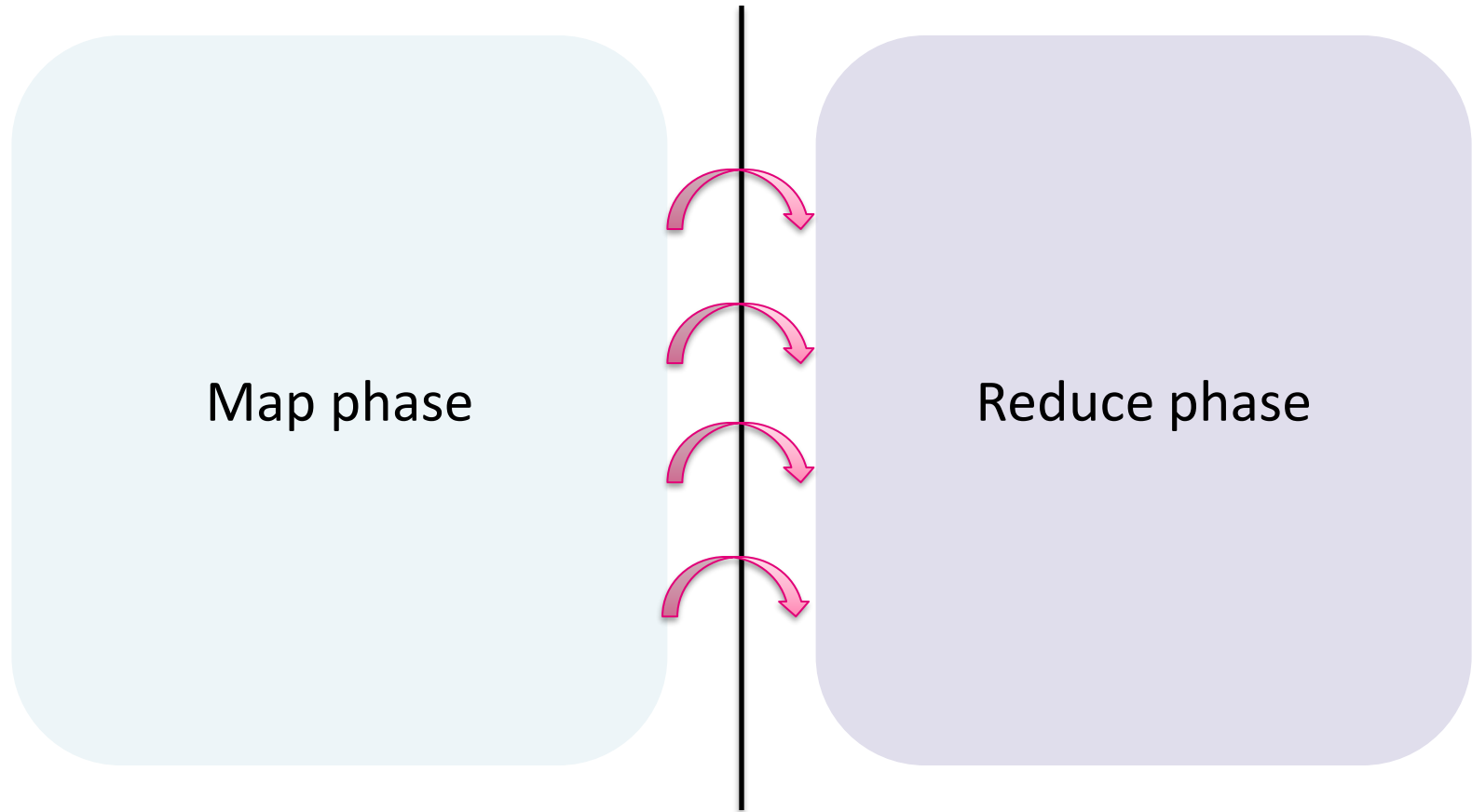
- Input: $(k1, v1)$
- Output: list $(k2, v2)$
- Independent for every key-value pair
→ mapping processes run in parallel

Shuffle Phase: intermediate results are shuffled through the network

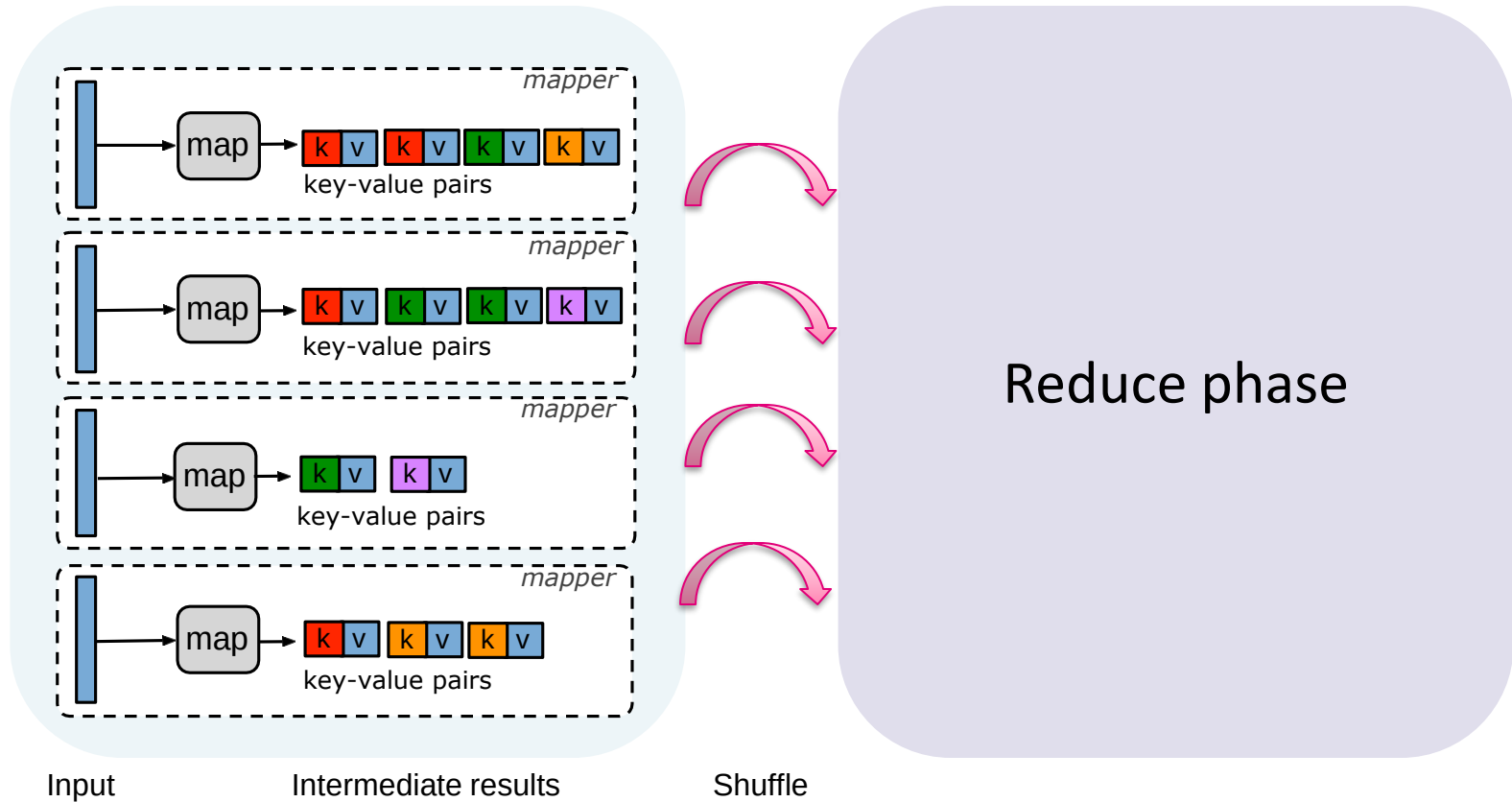
Reduce Phase

- Input: $(k2, \text{list}(v2))$
- Output: $(k2, v3)$
- Independent per group
→ reducer processes can run in parallel (per group)

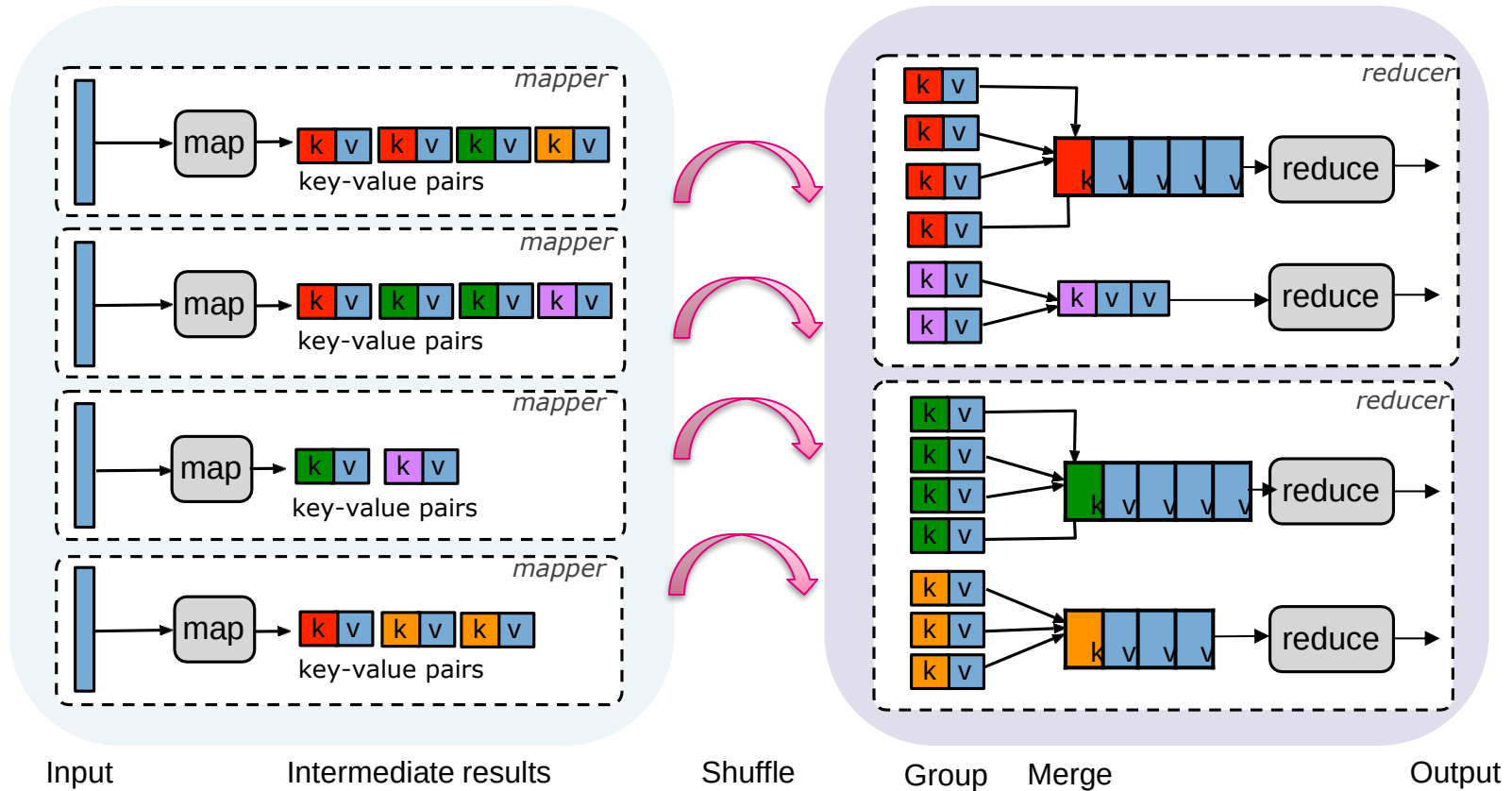
MapReduce Illustration



MapReduce Illustration



MapReduce Illustration



MapReduce Wordcount Example

Wordcount Example

- Given a text document, output the count for each word.

MapReduce interface

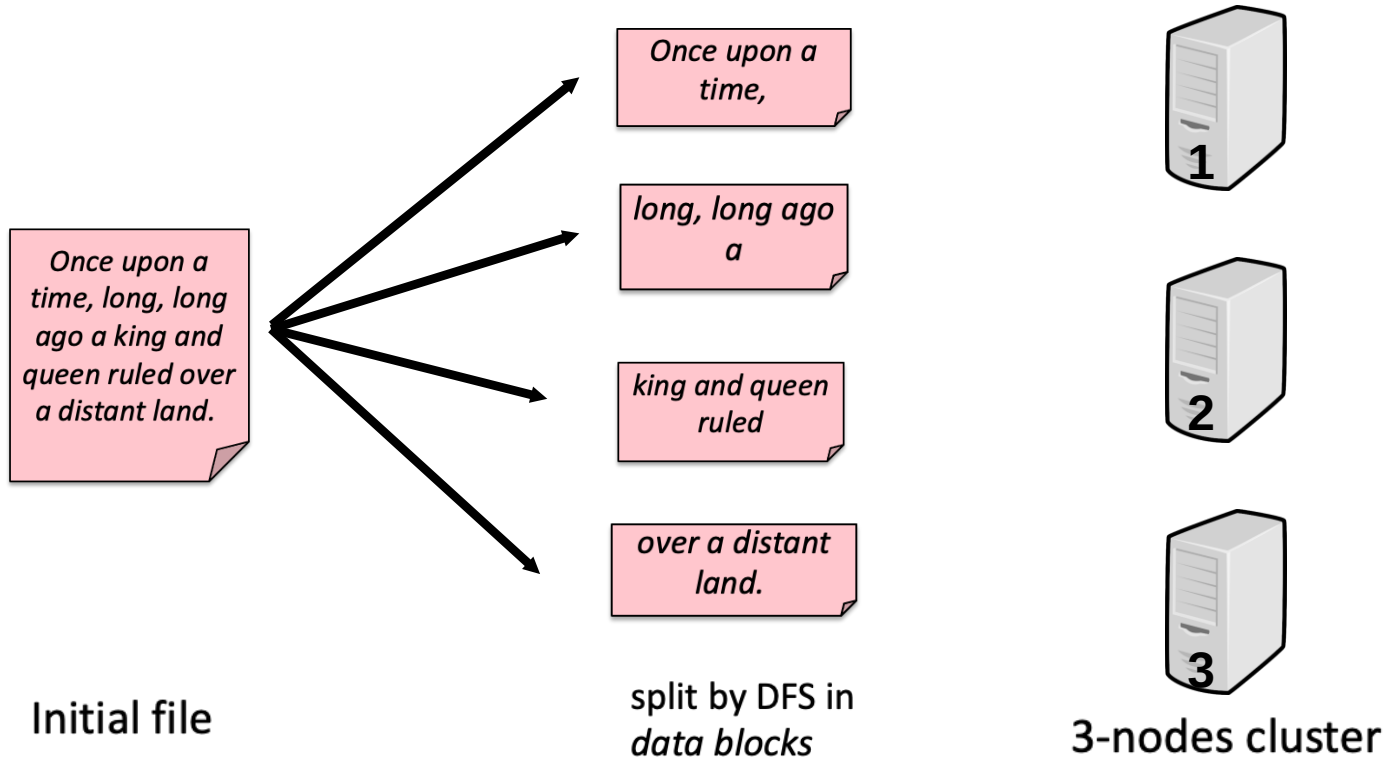
- Map $(k1, v1) \rightarrow \text{list}(k2, v2)$
- Reduce $(k2, \text{list}(v2)) \rightarrow (k2, v3)$

MapReduce Wordcount Example

MapReduce implementation

- Map (filename, text) $\rightarrow$ {word, 1}
- Reduce (word, [1,1,...]) $\rightarrow$ (word, count)

MapReduce Wordcount Example



MapReduce Wordcount Example

*Once upon a
time, long, long
ago a king and
queen ruled over
a distant land.*

Initial file

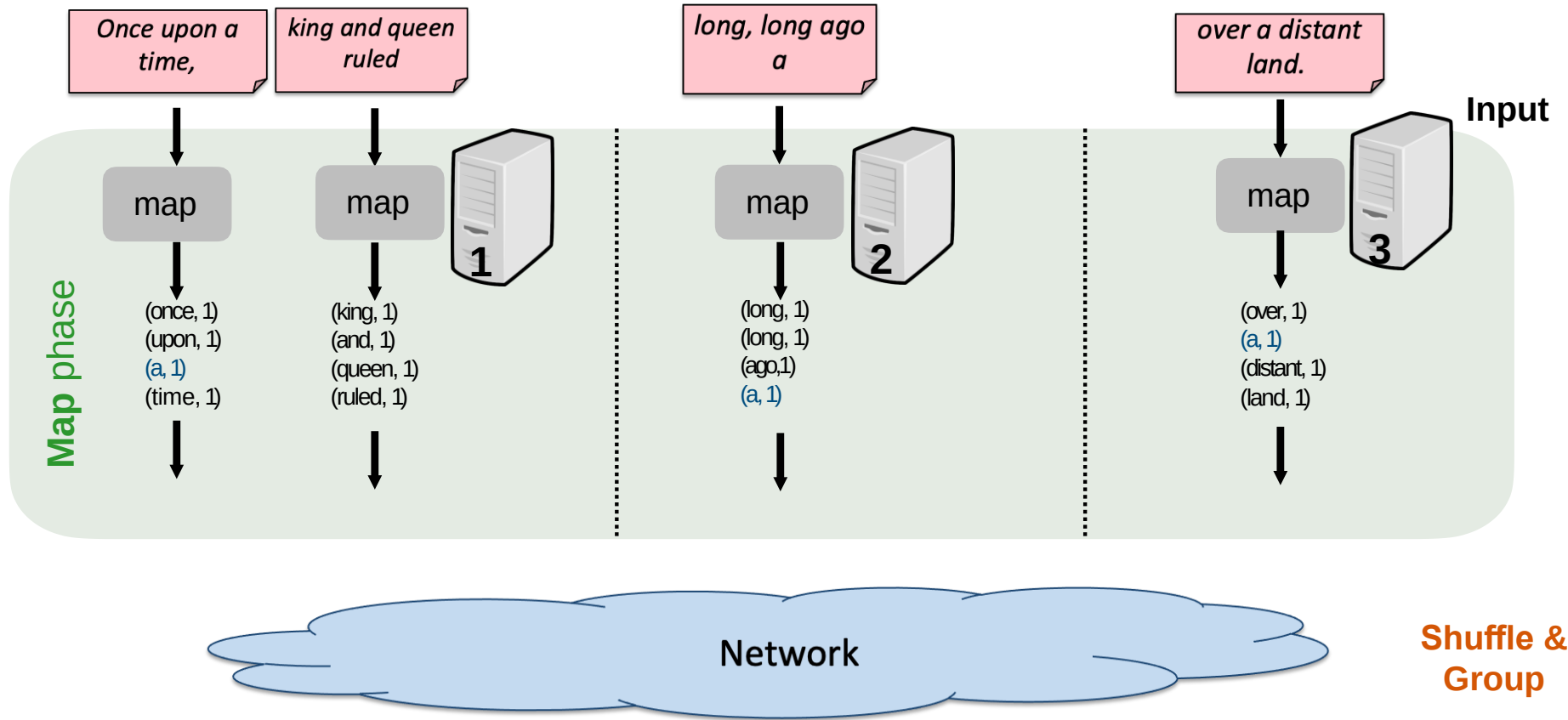
*Once upon a
time,
king and queen
ruled*

*long, long ago
a*

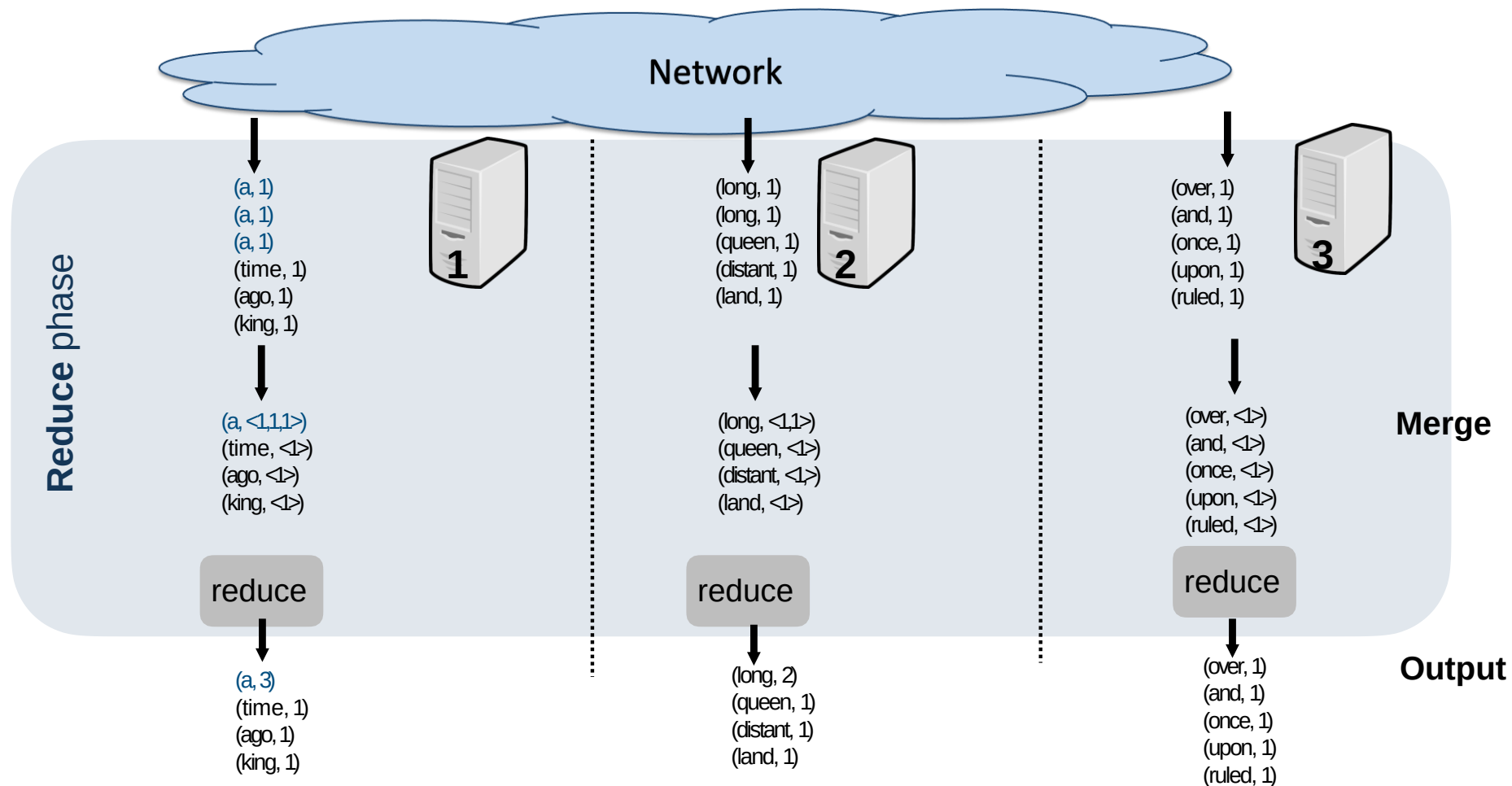
*over a distant
land.*

3-nodes cluster

Map Phase



Reduce Phase



Pseudocode

```
map(String key, String value):
```

?

```
reduce(String key, Iterator values):
```

?

Pseudocode

MapReduce

```
map(String key, String value):  
    // key: document name  
    // value: document contents  
    for each word w in value:  
        EmitIntermediate(w, "1");  
  
reduce(String key, Iterator values):  
    // key: a word  
    // values: a list of counts  
    int result = 0;  
    for each v in values:  
        result += ParseInt(v);  
    Emit(AsString(result));
```

Source: [Dean, Ghemawat; MapReduce: Simplified Data Processing on Large Clusters; OSDI'2004](#)

SQL

```
select word, count(*) from docs  
group by word
```

MapReduce Summary

Early tool to operate on Big Data

- Invented by Google in 2003
- Initially great tool for distributed processing of large amounts of data
- Used to build the web page index, many other applications
- Google moved away in 2010, killed it in 2014
- Apache Hadoop's market share on significant decline

Classes of Big Data Processing Systems

MapReduce Systems

Strengths

- Distributed storage
- Great scalability
- Able to process great variety of data sets (structured, semi-structured, unstructured)

Weaknesses



Menti.com code: 4332 3258



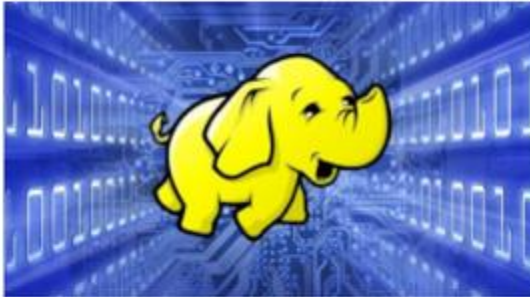
CLOUDERA



March 13, 2017

Hadoop Has Failed Us, Tech Experts Say

Alex Woodie



The Hadoop dream of unifying data and compute in a distributed manner has all but failed in a smoking heap of cost and complexity, according to technology experts and executives who spoke to *Datanami*.

“I can’t find a happy Hadoop customer. It’s sort of as simple as that,” says Bob Muglia, CEO of [Snowflake Computing](#), which develops and runs a cloud-based relational data warehouse offering. “It’s very clear to me, technologically, that it’s not the technology base the world will be built on going forward.”

Why bother?

Teaches foundational concepts in distributed systems

- **Parallel computing:** MapReduce divides data processing tasks into small, parallelizable units
- **Data processing:** Easy to understand data flow through the system
- **Fault tolerance:** internal mechanism to recover from failures
- **Scalability:** demonstrates how to scale “horizontally” (across many servers)

Learning Outcome 6: Discuss the pros and cons of different classes of data systems for modern analytics and data science applications

- Characteristics of Big Data
- Classes of Big Data Processing Systems
- **Example:** MapReduce, **Spark**



Improvement over MapReduce, developed in 2012

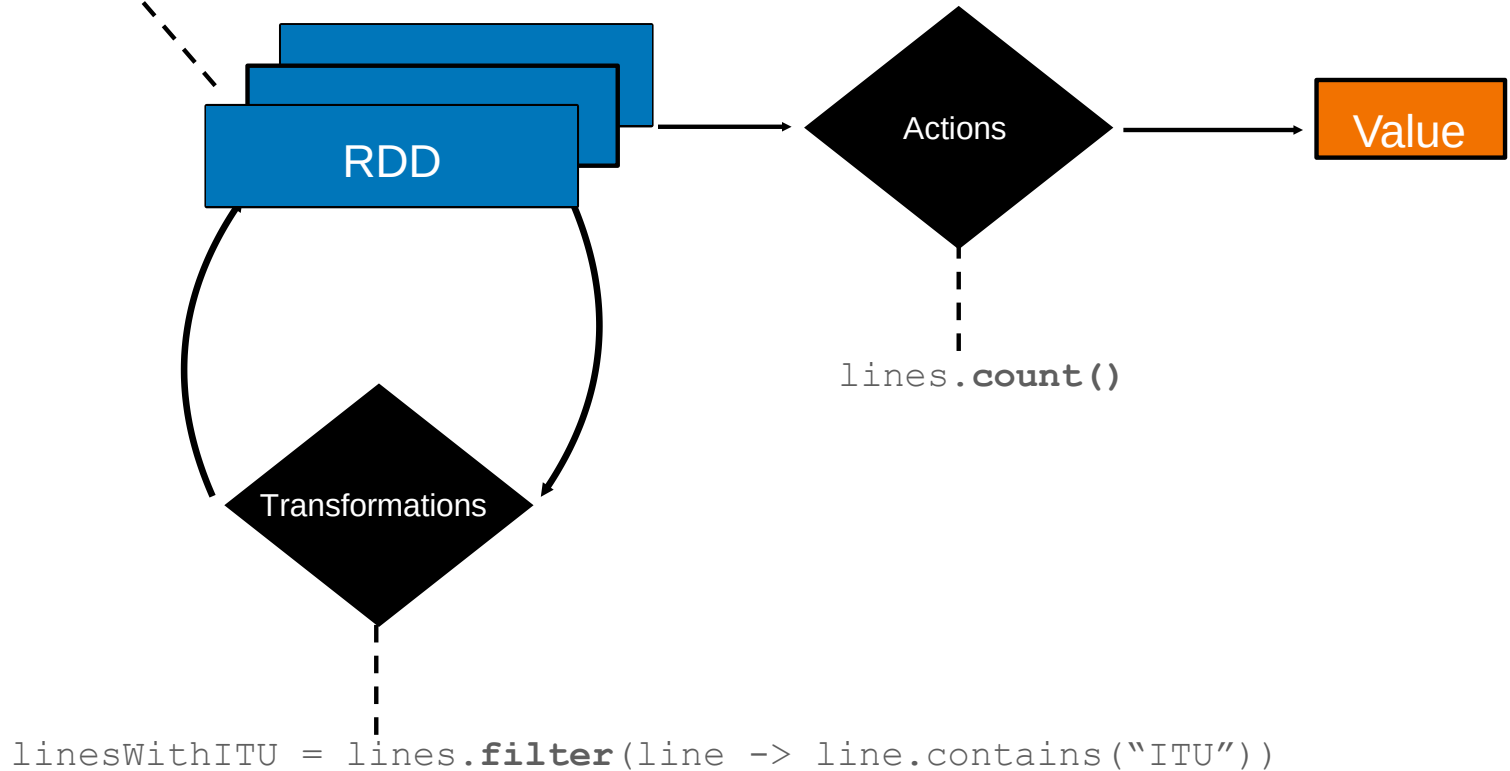
Data model: Resilient Distributed Datasets (RDDs)

- Transform one RDD to another via operators
- Lazy execution – optimizations

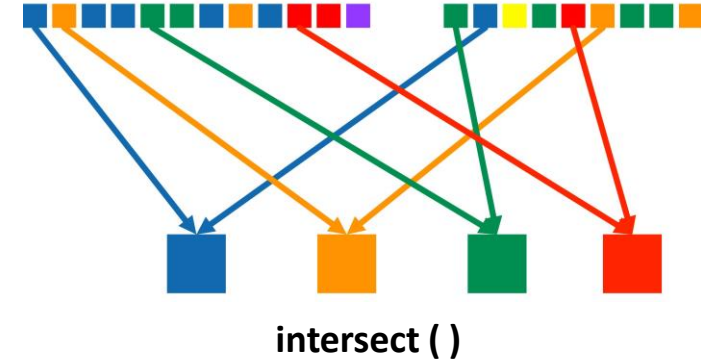
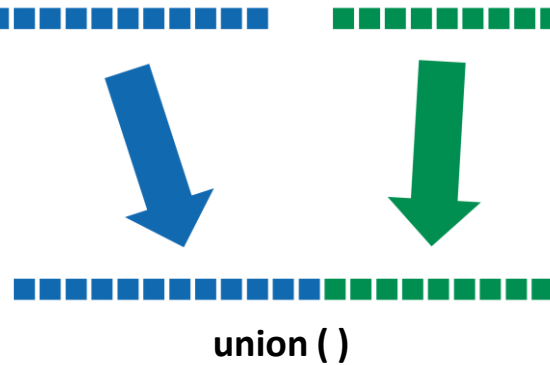
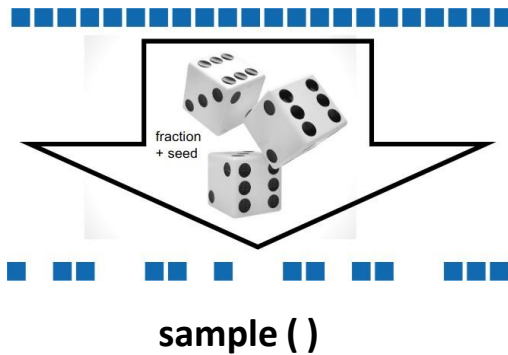
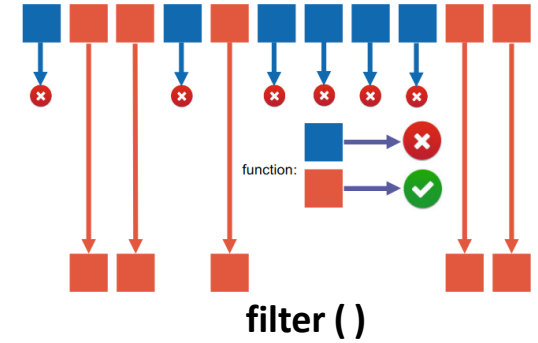
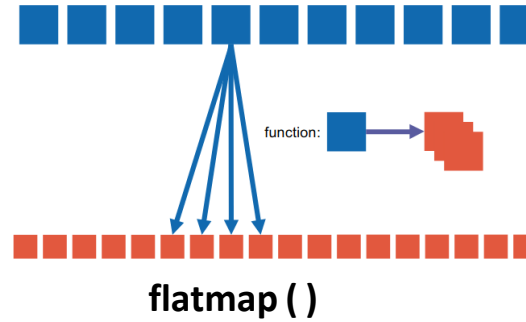
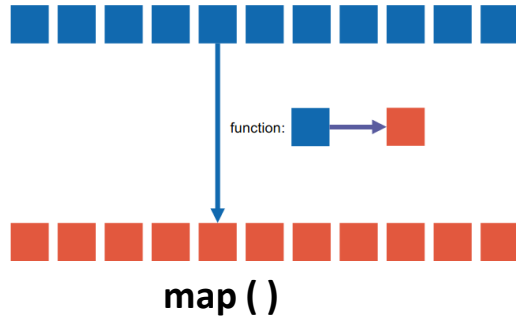
Write programs in terms of distributed datasets and operations on them

Working with RDDs

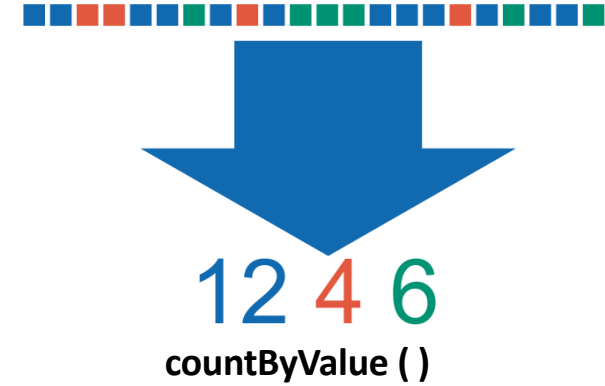
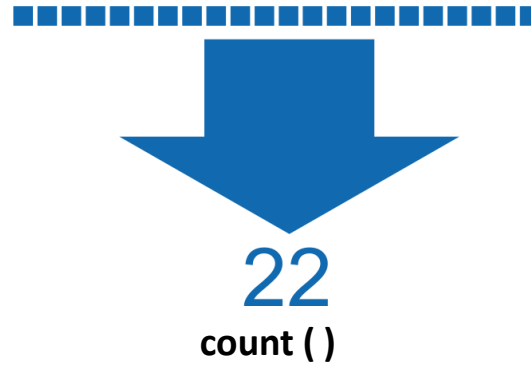
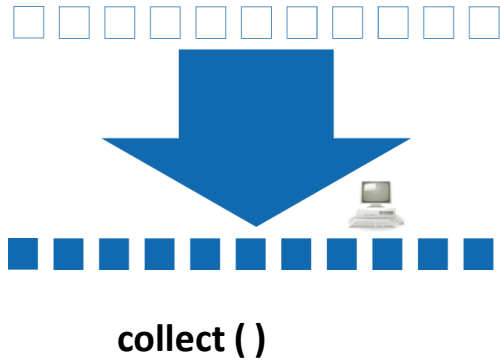
```
lines = sc.textFile("hdfs://data.txt")
```



Example Transformations

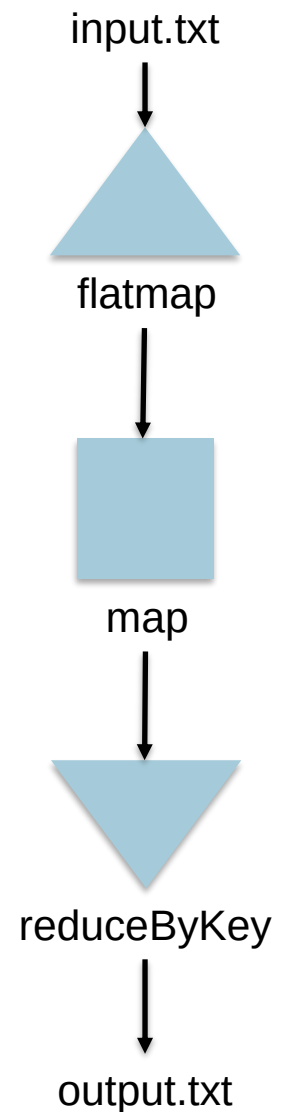


Example Actions



Spark Wordcount Example

```
file = sc.textFile("hdfs://...")
counts = file.flatMap(lambda line: line.split(" "))
               .map(lambda word: (word, 1))
               .reduceByKey(lambda a, b: a + b)
counts.saveAsTextFile("hdfs://...")
```



Learning Outcome 6: Discuss the pros and cons of different classes of data systems for modern analytics and data science applications

- Data sizes and requirements on systems continue to grow
- Many systems to process big data initially built to solve difficult problems (e.g. MapReduce for large, distributed data manipulation)
- SQL systems mostly caught up, pushed specialized systems out
- Pattern repeating (LLM training, vector databases)

Optional, but interesting reading material

- Michael Stonebraker, Andrew Pavlo, What Goes Around Comes Around... And Around..., SIGMOD Record, June 2024
<https://db.cs.cmu.edu/papers/2024/whatgoesaround-sigmodrec2024.pdf>